

Rockborne

Simple Linear Regression

Fitting a straight line to predict a number

Objectives



Intro to Simple LR

What it is and when to use it.



Establishing Variables

Dependent and independent variables.



The Equation

Slope, intercept and the error term.

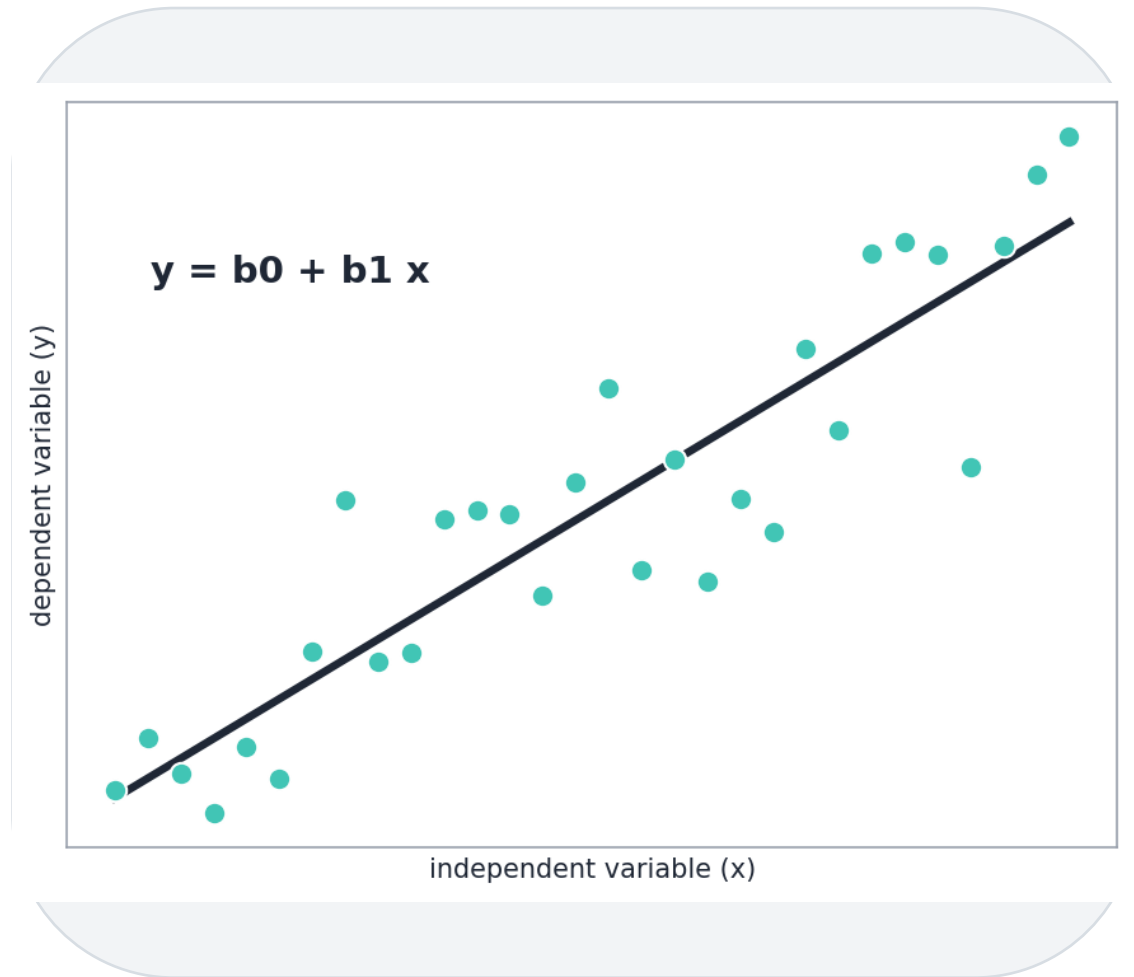
Introduction

Intro to Simple Linear Regression

...and why that matters

What is Simple Linear Regression?

- A simple equation that establishes a linear relationship between an input and a continuous numerical output.
- It can be used to predict a stock price, house prices, or customer lifetime value.
- One input, one output: the simplest regression.



Pros and Cons

Pros

- Easy to follow method.
- Results are ready for interpretation.
- Little to no learning curve vs other ML methods.

Cons

- Assumes a linear relationship.
- Handles outliers poorly.
- Risk of under-fitting small, high-dimensional data.

Simple Linear Regression



Establishing Variables

Dependent vs Independent Variables

Dependent variable (y)

- The target we want to explain or forecast.
- Its value changes based on the independent variable.
- Examples: test scores, blood sugar.

Independent variable (x)

- Impacts the dependent variable's output.
- Is not reactive to the dependent variable.
- Examples: time, age.

Simple Linear Regression



Implementing the Equation

The Equation: $y = b_0 + b_1 x + e$



b₀ - Intercept

The value of y when x = 0.



b₁ - Slope

How much y changes for each unit of x.



e - Error term

The residual: the gap between prediction and reality.

The Line of Best Fit



The goal

Find the line that sits closest to all the points.



OLS

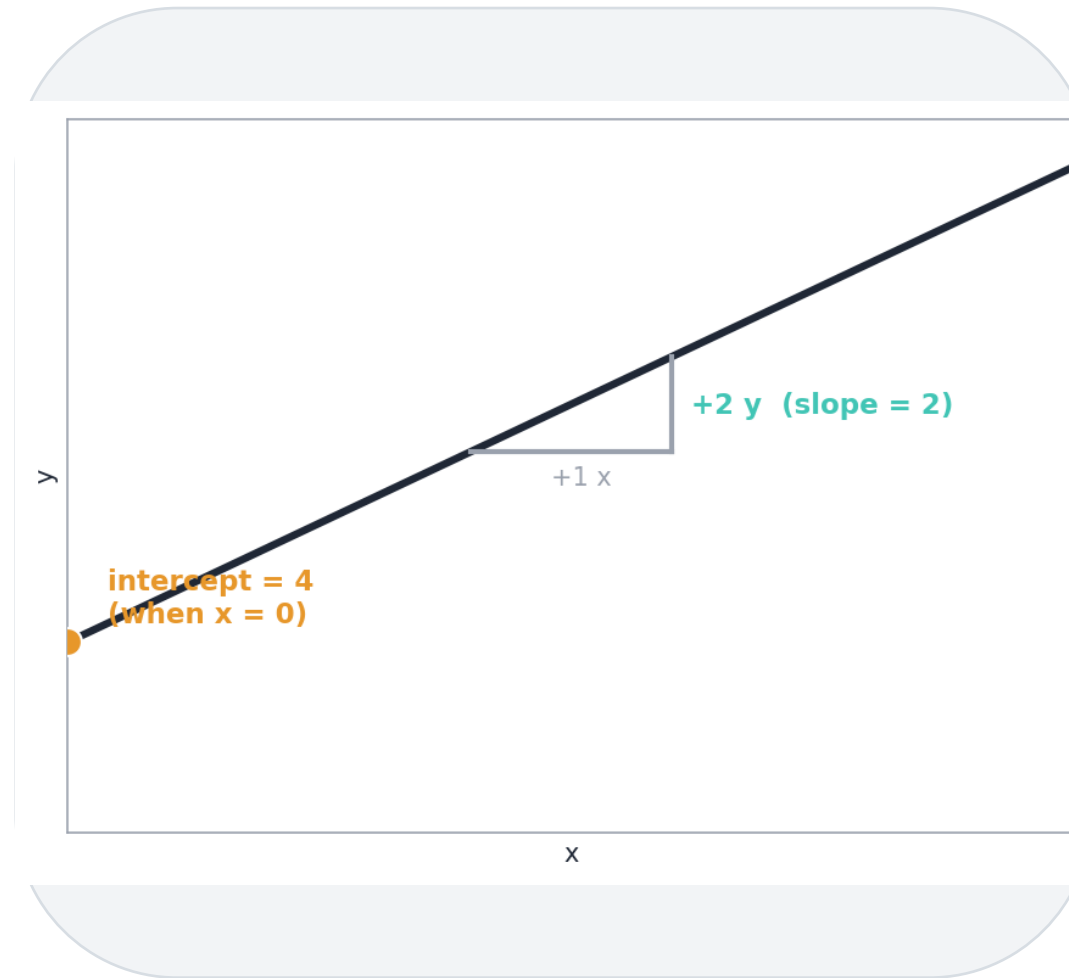
Ordinary least squares minimises the sum of squared residuals.



The result

The line with the smallest total error.

Slopes and Intercepts



1

From symbols to numbers: $y = 4 + 2x$.

2

The intercept is 4: when $x = 0$, $y = 4$.

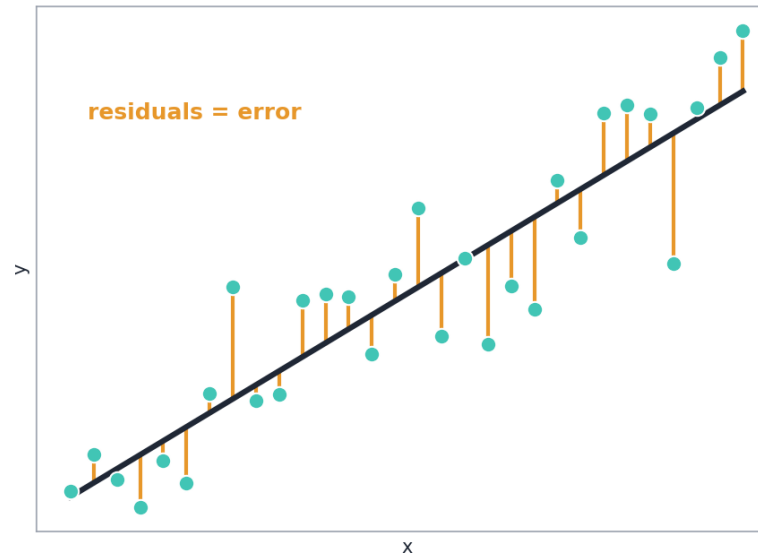
3

The slope is 2: when x increases by 1, y increases by 2.

4

The slope is the effect of the feature on the target.

The Error Term (Residuals)



The world is rarely perfectly linear.

The true values are the points; the line is the prediction.

The gaps between them are the residuals, the error.

OLS chooses the line that makes these residuals as small as possible.

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Time to get hands-on in Python

We move from theory to practice with a guided walkthrough you run yourself in Python.



Simple Linear Regression - Guided Walkthrough



Then: Multiple Linear Regression

In this lesson, we explored



Intro to Simple LR

A straight line linking one input to a numeric output.



Establishing Variables

Dependent (y) vs independent (x).



The Equation

Slope, intercept, and minimising residuals.

Digging Deeper?



[Linear Regression, real-life example](#)

Vitalflux



[What is Linear Regression?](#)

IBM



[Simple Linear Regression](#)

Statology

Questions?